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# Clinical Outcomes of Rapid Respiratory Virus Testing in Emergency Departments
## A Systematic Review and Meta-Analysis

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**IMPORTANCE** Rapid tests for respiratory viruses, including multiplex panels, are increasingly available in emergency departments (EDs). Their association with patient outcomes remains unclear.

**OBJECTIVE** To determine if ED rapid respiratory virus testing in patients with suspected acute respiratory infection (ARI) was associated with decreased antibiotic use, ancillary tests, ED length of stay, and ED return visits and hospitalization and increased influenza antiviral treatment.

**DATA SOURCES** Ovid MEDLINE, Embase (Ovid), Scopus, and Web of Science from 1985 to November 14, 2022.

**STUDY SELECTION** Randomized clinical trials of patients of any age with ARI in an ED. The primary intervention was rapid viral testing.

**DATA EXTRACTION AND SYNTHESIS** Preferred Reporting Items for Systematic Reviews and Meta-Analyses reporting guidelines were followed. Two independent reviewers (T.S. and K.W.) extracted data and assessed risk of bias using the Cochrane Risk of Bias, version 2.0. Estimates were pooled using random-effects models. Quality of evidence was assessed using the Grading of Recommendations, Assessment, Development, and Evaluations framework.

**MAIN OUTCOMES AND MEASURES** Antibiotic use and secondary outcomes were pooled separately as risk ratios (RRs) and risk difference estimates with 95% CIs.

**RESULTS** Of 7157 studies identified, 11 (0.2%; n = 6068 patients) were included in pooled analyses. Routine rapid viral testing was not associated with antibiotic use (RR, 0.99; 95% CI, 0.93-1.05; high certainty) but was associated with higher use of influenza antivirals (RR, 1.33; 95% CI, 1.02-1.75; moderate certainty) and lower use of chest radiography (RR, 0.88; 95% CI, 0.79-0.98; moderate certainty) and blood tests (RR, 0.81; 95% CI, 0.69-0.97; moderate certainty). There was no association with urine testing (RR, 0.95; 95% CI, 0.77-1.17; low certainty), ED length of stay (0 hours; 95% CI, −0.17 to 0.16; moderate certainty), return visits (RR, 0.93; 95%, CI 0.79-1.08; moderate certainty) or hospitalization (RR, 1.01; 95% CI, 0.95-1.08; high certainty). Adults represented 963 participants (16%). There was no association of viral testing with antibiotic use in any prespecified subgroup by age, test method, publication date, number of viral targets, risk of bias, or industry funding.

**CONCLUSIONS AND RELEVANCE** The results of this systematic review and meta-analysis suggest that there are limited benefits of routine viral testing in EDs for patients with ARI. Further studies in adults, especially those with high-risk conditions, are warranted.

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Supplemental content

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Acute respiratory tract infections (ARIs) are the most common cause of medically attended acute illness.[1] Clinically, it is difficult to distinguish bacterial etiologies or influenza (for which specific treatments are available) from ARI caused by other respiratory viruses. This diagnostic uncertainty is associated with unnecessary antibiotic treatment and subsequent adverse drug events, increased health care costs, and antibiotic resistance.[2] Accordingly, some antimicrobial stewardship guidelines advocate for rapid viral (RV) testing for respiratory viruses to decrease use of antibiotics.[3]

The SARS-CoV-2 pandemic was followed by increased availability of RV testing, including multiplex panels, in emergency departments (EDs).[4] However, the association of these tests with patient outcomes is unclear. Previous meta-analyses that included studies until 2017 showed that RV testing in ambulatory care was associated with a reduction of antibiotic prescribing in observational studies, but not in randomized clinical trials (RCTs).[5] Substantial new RCT data investigating the effect of molecular multiplex panels warrants a new assessment.

This systematic review and meta-analysis aimed to determine if the use of rapid respiratory viral diagnostic testing in patients of all ages presenting in the ED for ARI was associated with decreased ED antibiotic prescribing and other clinically relevant outcomes. These included the use of influenza antivirals, ancillary testing, ED length of stay, ED return visits, or hospitalization.

### Key Points

**Question** Is rapid testing for respiratory viruses associated with patient treatment in the emergency department (ED)?

**Findings** In this systematic review and meta-analysis of 11 randomized clinical trials, rapid viral testing was not associated with reduced antibiotic use, ED length of stay, and the rate of ED return visits or of hospitalization. However, rapid viral testing was associated with moderately increased influenza antiviral use (absolute risk difference 1%) and decreased use of chest radiography and blood tests (absolute risk difference, 3%-4% each).

**Meaning** The results of this meta-analysis suggest that the benefits of ED rapid viral testing are limited for the general population.

## Methods

The protocol was developed according to the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) statement and registered with the international prospective register of systematic reviews (PROSPERO; CRD42018103672). Reporting followed PRISMA.

### Information Sources and Search Strategy

We developed an electronic search strategy (eMethods 2 in Supplement 1) in collaboration with a medical librarian and searched Ovid MEDLINE, Embase (Ovid), Scopus, and Web of Science Core Collection for studies published after January 1, 1985. To identify additional studies, we screened the reference lists of included studies and relevant reviews. The review was initially planned to include observational studies and RCTs. A first search was performed on June 1 to June 4, 2018. During full-text screening, we identified a sufficient number of RCTs and amended the protocol to limit the analysis to RCTs. The search was updated on November 14, 2022.

### Study Design and Participants

We included published original peer-reviewed full reports of RCTs that evaluated the clinical effect of the routine use of respiratory virus testing for physician decision-making in the ED. Our definition of RCT included full RCTs (using patient-level randomization), and quasi-RCTs (ie, those using a quasirandom method of allocation, such as alternating days). Included studies assessed patients of any age presenting to an ED with ARI. ARI was defined as a new illness with respiratory symptoms suggestive of infection. Studies restricted to populations with specific chronic health conditions were excluded.

The primary intervention was availability of rapid respiratory virus testing (defined as the provision of test results during the patient's ED stay) or the awareness of the treating physician of the rapid test results. Secondary intervention was RV test positivity vs negativity. The primary outcome was the association with antibiotic prescription during the ED visit. Secondary outcomes were influenza antiviral use, ancillary testing (including chest radiography, blood culture, urinalysis or urine culture, and any blood test), ED length of stay, ED return visits, or hospitalization. When parts of composite outcomes (ie, urinalysis or urine culture; blood culture or other blood test) were reported individually, the variable with the higher number of events was chosen. Additionally, we determined which social determinants of health were captured using the PROGRESS Plus framework.[6,7]

### Study Selection

Articles were screened by 1 reviewer (T.S.) at title and abstract level. Full-text screening, data extraction, and quality assessment using the Cochrane Risk-of-Bias tool RoB-2[8] were done independently by 2 reviewers (T.S. and K.W.). Discussion or a third reviewer resolved conflicts. Screening and data extraction were performed using DistillerSR (DistillerSR Inc). Corresponding authors were contacted for missing information.

### Statistical Analysis

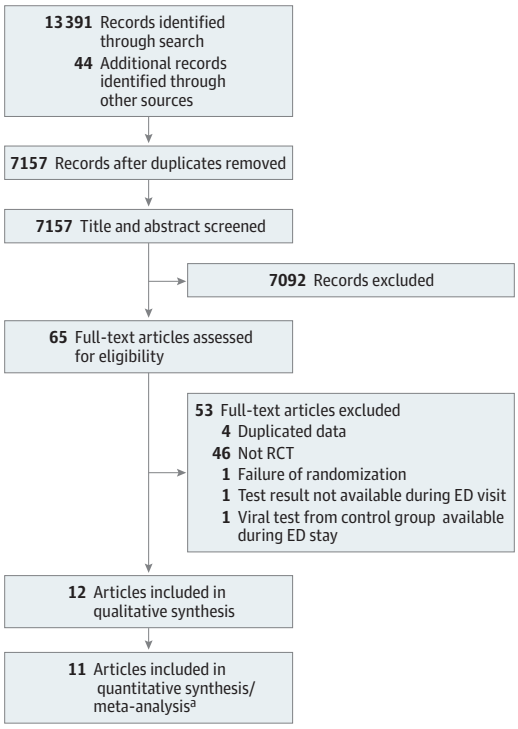
Outcomes for each of the 2 interventions were analyzed and presented separately. Associations of the intervention with dichotomous outcomes were expressed as risk ratios (RRs) and risk differences (RDs) with 95% CIs, and continuous outcomes were expressed as standardized mean differences with 95% CIs. If 2 or more studies were available, we performed meta-analyses using a random-effects model with the restricted maximum likelihood method. Statistical heterogeneity was assessed using the $I^2$ statistic. We conducted meta-analyses within the following prespecified strata: children and



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Figure 1. PRISMA Flow Diagram of Included and Excluded Articles




[a] One study not included as it only reported changes in proportional management without providing absolute numbers. ED indicates emergency department; RCT, randomized clinical trial.

adolescents vs adults, antigen detection (enzyme immunoassay [EIA] or immunofluorescence) vs molecular tests, monoplex (influenza) vs multiplex tests, low vs high risk of bias, and industry funding vs no industry funding. Differences in pooled RRs between subgroups were assessed via fixed-effects meta-regression models in which the subgroup of interest was included as a covariate. All tests were 2-sided with a significance level of .05. Analyses were conducted using R, version 4.2.2 (R Foundation for Statistical Computing); the metafor package, version 4.0-0 (Microsoft); and Excel 2016 (Microsoft). Narrative summaries were presented for results that could not be meta-analyzed.

### Certainty of Evidence

Certainty of evidence was assessed by 2 reviewers for all outcomes of the primary exposure using the Grading of Recommendations Assessment, Development and Evaluation (GRADE).[9,10] We rated certainty of evidence as very low, low, moderate, or high based on risk of bias,[8] imprecision, inconsistency, indirectness, and publication bias. Outcomes in which most studies were judged as low risk of bias were considered to be overall of low risk of bias.[9] We chose a minimally contextualized approach to rate imprecision following the guidance of the GRADE Working Group.[11,12] Minimal important differences (MID) were set at 10% for all outcomes based on previous studies on outpatient antibiotic use and clinical judgment.[13] When the magnitude of pooled estimates were less than ±10%, we rated certainty as little or no association; otherwise, we rated certainty in showing an association with the MID as a threshold. We rated down for imprecision by 2 levels when the 95% CI crossed more than 1 threshold of importance. Publication bias was assessed for meta-analyses by the Egger test and funnel plots for groups with 10 or more studies.[14]

## Results

Of 7157 publications identified, 7092 (99.1%) were excluded based on title and abstract screening and the remaining 65 (0.9%) underwent full-text review. After exclusion of 53 studies (eMethods in Supplement 1), 12 were included (**Figure 1**).

### Study Characteristics

Of the 12 included studies, 5 were quasi-RCTs[15-19] and 7 were full RCTs[20-26] (**Table 1**[15-26]). In the intervention arm, 4 studies tested for influenza only,[17,18,20,23] and the other 8 tested for multiple respiratory viruses. Four studies (all published before 2010) used EIAs,[17,18,20,23] 1 used immunofluorescence,[22] and 7 studies (all published since 2017) used molecular testing.[15,16,19,21,24-26] Multiplex tests included influenza and respiratory syncytial virus,[15] influenza/respiratory syncytial virus/adenovirus/parainfluenza 1 to 3,[22] or a panel of 15 or more respiratory viruses.[16,19,21,24-26] No study evaluated testing for SARS-CoV-2.

Comparators varied: the control groups in the influenza-only trials did not test for any viruses,[17,18,23] or the treating physician was unaware of the result.[20] In the rapid multiplex testing studies, the comparator was either multiplex testing in a central laboratory with a longer turnaround time,[15,16] immunofluorescence,[19] or routine care that included laboratory-based viral testing at the treating physician's discretion.[21,22,24-26]

The age of participants varied, but pediatric populations were dominant. Eight studies were limited to children and adolescents,[17-20,22-24,26] including 4 studies in children 6 years or younger.[17-19,22] Two studies included adults and children.[15,25]

### Quality of Included Studies

All quasi-RCTs were judged at high risk of bias, and 7 of the 8 RCTs were judged at low risk of bias (eFigure 1 in Supplement 1). The RCT by Echavarria et al[19] was at a high risk of bias due to deviation from the intended intervention, as less than half of patients were randomized as planned. None of the studies were able to mask participants and personnel to testing or test results. No study masked outcome assessors to test status.

### Antibiotic Use

Antibiotic prescription during the ED visit was reported in all included studies. The study by Echavarria et al[19] could not be included in meta-analyses as it only reported changes in proportional management without providing absolute numbers. Accordingly, 11 studies were meta-analyzed, which showed with high certainty of evidence little or no difference in antibiotic use between RV testing and controls (RR, 0.99; 95% CI,



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Table 1. Characteristics of Included Studies

| Study | Study design | No. of patients | Age range | Setting; country | Target and type of rapid test | Rapid test | Comparator |
|---|---|---|---|---|---|---|---|
| Bonner et al,[20] 2003 | RCT | 391 | 2 mo to 21 y | ED; US | Influenza; antigen | FluOIA Biostar | Same test, result unknown |
| Esposito et al,[23] 2003 | RCT | 957 | 0-15 y | ED; Italy | Influenza; antigen | Sofia Quickvue | No test |
| Iyer et al,[17] 2006 | Quasi-RCT (alternating days) | 700 | 2-24 mo | ED; US | Influenza; antigen | Sofia Quickvue | No test |
| Poehling et al,[18] 2006 | Quasi-RCT (randomized days) | 305 | <5 y | ED; US | Influenza; antigen | Sofia Quickvue | No test |
| Doan et al,[22] 2009 | RCT | 199 | 3-36 mo | ED; Canada | Multiple; immunofluorescence[a] | SimulFluor | Routine care |
| Brendish et al,[21] 2017 | RCT | 279 | ≥18 y | ED and acute medical unit; UK[b] | Multiple; molecular | BioFire FilmArray | Routine care |
| Echavarría et al,[19] 2018[c] | RCT | 432 | 2 mo to 6 y | ED; Argentina | Multiple; molecular[d] | BioFire FilmArray | Immunofluorescence |
| May et al,[25] 2019 | RCT | 191 | ≥12 mo[e] | ED; US | Multiple; molecular[d] | BioFire FilmArray | Routine care |
| Bouzid et al,[16] 2021 | Quasi-RCT (alternating weeks) | 474 | ≥18 y | ED; France | Multiple; molecular[f] | QIAstat-Dx | Respiratory panel in centralized laboratory |
| Rao et al,[26] 2021 | RCT | 908 | 1 mo to 18 y | ED; US | Multiple; molecular[d] | BioFire FilmArray | Routine care |
| Bibby et al,[15] 2022 | Quasi-RCT (alternating days) | 421 | All age groups[e] | ED and inpatients; Canada[b] | Influenza and RSV; molecular | Xpert Xpress | Respiratory panel in centralized laboratory |
| Matilla et al,[24] 2022 | RCT | 1243 | 0-17 y | ED; Finland | Multiple; molecular[f] | QIAstat-Dx | Routine care |

Abbreviations: ED, emergency department; RCT, randomized clinical trial; RSV, respiratory syncytial virus.

[a] Includes adenovirus, influenza, parainfluenza 1 to 3, and RSV.

[b] Only ED data analyzed for the current systematic review.

[c] Not included in any meta-analysis except for the hospitalization outcome.

[d] Includes adenovirus; coronaviruses HKU1, NL63, 229E, and OC43; human metapneumovirus; influenza; rhinovirus/enterovirus; RSV; and parainfluenza 1 to 4.

[e] Separate data for children and adults available.

[f] Includes adenovirus; bocavirus; coronaviruses HKU1, NL63, 229E, and OC43; human metapneumovirus; influenza; rhinovirus/enterovirus; RSV; and parainfluenza 1 to 4.

0.93-1.05; $I^2$ = 0.03%; **Table 2**[15-26]; **Figure 2**A). A funnel plot and Egger test did not suggest publication bias (eFigure 2 in Supplement 1). Overall prevalence of antibiotic use differed substantially between studies, ranging from 8.5% to 61.9% (mean [SD], 26.3% [14.7%]) in children and 18.9% to 76.7% (mean [SD], 46.5% [25.8%]) in adults. There was no association of viral testing with antibiotic use in any of the following prespecified subgroup analyses: children and adolescents, adults, antigen-based testsor immunofluorescence, molecular tests (which also corresponded to the more recently published studies), monoplex tests, multiplex tests, low risk of bias, high risk of bias, industry funding, and no industry funding (**Table 3**[15-26]).

Seven studies also evaluated antibiotic use according to the test result (positive vs negative).[15,17,20-23] Overall, patients with a positive viral test result in the RV testing group were less likely to receive antibiotics (RR, 0.53; 95% CI, 0.37-0.77; $I^2$ = 65.7%; Figure 2B) than those with a negative result. Antibiotic use in the virus-negative RV testing group was correspondingly higher compared with the virus-positive group and also compared with the corresponding control group without RV testing (52.8% vs 38.5%; $P$ = .03). Subgroup analyses demonstrated lower rates of antibiotic use among virus-positive cases for traditional tests, monoplex tests, low risk of bias, and high risk of bias, but not for molecular and multiplex tests (eTable 1 in Supplement 1).

Antibiotic duration depending on test availability was reported by 2 articles to be comparable in the groups with and without RV testing, with a median of 7 vs 7 days[23] and 6.8 vs 6.5 days, respectively.[21] Two studies reported whether patients received antibiotics at follow-up within 7 days after discharge from the ED. There was no difference in the study by Matilla et al[24] (34.1% vs 34.5%). In contrast, Doan et al[22] reported less antibiotic use at follow-up (5.6% vs 15.5%, RR, 0.36; 95% CI, 0.14-0.95).

## Antiviral Use

Influenza antiviral use was reported in 7 studies.[15,16,18,20,21,25,26] Meta-analysis showed an increase in antiviral prescribing with RV testing with moderate certainty of evidence (RR, 1.33; 95% CI, 1.02-1.75; $I^2$ = 0%; absolute RD, 1.4%; Table 2; eFigure 3 in Supplement 1). This association was significant in the 2 monoplex studies, which did not offer influenza testing in the control arm (RR, 2.12; 95% CI, 1.0-4.51; $I^2$ = 0%),[18,20] but not with multiplex testing (RR, 1.24; 95% CI, 0.93-1.66; $I^2$ = 0%).[15,16,21,25,26] The association with antiviral use did not differ significantly ($P$ = .85) between children (RR, 1.25; 95% CI, 0.77-2.03) and adults (RR, 1.18; 95% CI, 0.81-1.72). Six studies reported influenza antiviral use depending on the rapid influenza test result.[15,18,20,21,25,26] Mean influenza antiviral use per study was 23.2% for influenza-positive patients and 2.8%



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Table 2. Summary of Results for Rapid Viral Test Availability

| Outcome | No. of studies | No. of patients | Relative association estimate | Absolute association estimate | | Certainty of evidence (GRADE) | Plain language summary |
|---|---|---|---|---|---|---|---|
| | | | | Rapid viral testing | Control | | |
| Antibiotic use | 11[15-18,20-26] | 6068 | 0.99; 95% CI, 0.93 to 1.05; $I^2$ = 0.03% | 1111 per 3206; 34.7% | 1007 per 2862; 35.2% | High | There is little or no difference between rapid viral test and control in antibiotic use |
| | | | | Risk difference, −0.01; 95% CI, −0.04 to 0.02; $I^2$ = 43.4% | | | |
| Influenza antiviral use | 7[15,16,18,20,21,25,26] | 2969 | 1.33; 95% CI, 1.02 to 1.75; $I^2$ = 0% | 116 per 1465; 7.9% | 85 per 1504; 5.7% | Moderate[a] | Rapid viral testing probably increases influenza antiviral use |
| | | | | Risk difference: 0.01; 95% CI, 0 to 0.03; $I^2$ = 0% | | | |
| Chest radiography | 8[15,17,18,20,22-25] | 4408 | 0.88; 95% CI, 0.79 to 0.98; $I^2$ = 0% | 417 per 2346; 17.8% | 444 per 2062; 21.5% | Moderate[a] | Rapid viral testing probably decreases chest radiography use |
| | | | | Risk difference: −0.03; 95% CI, −0.05 to 0; $I^2$ = 31.1% | | | |
| Blood test (any) | 5[17,18,20,22,23] | 2552 | 0.81; 95% CI, 0.69 to 0.97; $I^2$ = 0% | 188 per 1240; 15.2% | 246 per 1312; 18.8% | Moderate[a] | Rapid viral testing may decrease blood testing |
| | | | | Risk difference: −0.04; 95% CI, −0.06 to −0.01; $I^2$ = 0% | | | |
| Blood culture | 2[17,20] | 1091 | 0.85; 95% CI, 0.67 to 1.07; $I^2$ = 0% | 95 per 538; 17.7% | 116 per 553; 21.0% | Very low[a,b,c] | It is uncertain whether rapid viral testing decreases blood culture testing |
| | | | | Risk difference: −0.03; 95% CI, −0.07 to 0.01; $I^2$ = 0% | | | |
| Blood test (other) | 4[17,20,22,23] | 2247 | 0.84; 95% CI, 0.70 to 1.01; $I^2$ = 0% | 174 per 1105; 15.7% | 215 per 1142; 18.8% | Low[a,b] | Rapid viral testing may decrease other blood testing |
| | | | | Risk difference: −0.03; 95% CI, −0.06 to 0; $I^2$ = 0% | | | |
| Urine analysis/culture | 4[17,18,20,22] | 1595 | 0.95; 95% CI, 0.77 to 1.17; $I^2$ = 0% | 130 per 762; 17.1% | 153 per 833; 18.4% | Low[a,c] | Rapid viral testing may have little or no association with urine testing |
| | | | | Risk difference: −0.02; 95% CI, −0.05 to 0.02; $I^2$ = 0% | | | |
| ED length of stay | 4[17,22,24,25] | 2333 | 1.02; 95% CI, 0.96 to 1.08; $I^2$ = 63.4%[d] | Mean (SD), 3.40 (1.78) | Mean (SD), 3.53 (1.96) | Moderate[e] | There is probably little or no difference between rapid viral test and control in ED length of stay |
| | | | | Standardized mean difference, 0; 95% CI, −0.17 to 0.16; $I^2$ = 67.4% | | | |
| ED return visit | 7[17,21,22,24-26] | 3520 | 0.93; 95% CI, 0.79 to 1.08; $I^2$ = 0% | 282 per 1941; 14.5% | 249 per 1579; 15.8% | Moderate[a] | There is probably little or no difference between rapid viral test and control in ED return visit |
| | | | | Risk difference: −0.01; 95% CI, −0.03 to 0.02; $I^2$ = 0% | | | |
| Hospitalization | 9[16-19,21,23-26] | 5489 | 1.01; 95% CI, 0.95 to 1.08; $I^2$ = 0% | 882 per 3029; 29.1% | 642 per 2460; 26.1% | High | There is little or no difference between rapid viral test and control in hospitalization rate |
| | | | | Risk difference: 0.00; 95% CI, −0.02 to 0.02; $I^2$ = 0% | | | |

Abbreviations: ED, emergency department; GRADE, Grading of Recommendations, Assessment, Development, and Evaluations.

[a] Rated down 1 level for imprecision because of the 95% CI crossing the minimal important difference decision threshold.

[b] Rated down 1 level for imprecision because of the 95% CI crossing the null effect threshold.

[c] Rated down 1 level for bias as more than half of included studies were high risk of bias during randomization process (ie, quasi-randomized clinical trials).

[d] Corresponding relative effect estimate log-transformed ratio of means: 1.02; 95% CI, 0.96-1.08; $I^2$ = 63.4%.

[e] Rated down 1 level due to heterogeneity/inconsistency.

for influenza-negative patients (RR, 9.8; 95% CI, 3.27-29.4; $I^2$ = 75.5%).

## Ancillary Tests

Eight studies reported on chest radiography.[15,17,18,20,22-25] Meta-analysis showed lower chest radiography use among patients with RV testing with moderate certainty (RR, 0.88; 95% CI, 0.79-0.98; $I^2$ = 0%; eFigure 4 in Supplement 1), corresponding to an absolute RD of 2.6%. RV testing was associated with decreased blood testing with moderate certainty (RR, 0.81; 95% CI, 0.69-0.97; $I^2$ = 0; absolute RD, 3.7%; eFigure 5 in Supplement 1).[15,17,18,22-25] Differentiation of blood testing into blood cultures (RR, 0.85; 95% CI, 0.67-1.07; $I^2$ = 0) and other blood tests (RR, 0.84; 95% CI, 0.70-1.01; $I^2$ = 0%) demonstrated possible reductions in testing (very low and low certainty, respectively). In contrast, RV testing appeared to have little or no association with urine testing (RR, 0.95; 95% CI, 0.77-1.17; $I^2$ = 0%; low certainty; eFigure 6 in Supplement 1). Among studies also reporting on the use of ancillary tests depending on RV test results, all examined ancillary tests were performed less frequently among patients with a positive viral test result, with RRs between 0.2 and 0.47 (eTable 2 in Supplement 1).

## Additional Outcomes

The association of RV testing with ED length of stay was available in 6 studies. We could perform a meta-analysis of 4 studies[17,22,24,25] that reported means and standard deviations.



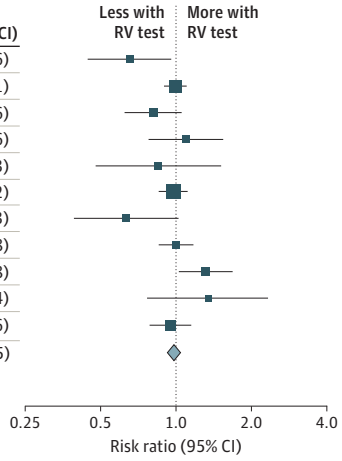
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Figure 2. Association of Rapid Viral (RV) Testing and a Positive vs Negative RV Test Result With Antibiotic Use

A RV test

| Study | RV test Antibiotic | RV test Total | Control Antibiotic | Control Total | Risk ratio (95% CI) | Weight, % |
|---|---|---|---|---|---|---|
| Bonner et al,[20] 2003 | 34 | 193 | 53 | 198 | 0.66 (0.45-0.96) | 2.38 |
| Esposito et al,[23] 2003 | 296 | 478 | 296 | 479 | 1.00 (0.91-1.11) | 35.12 |
| Iyer et al,[17] 2006 | 77 | 345 | 97 | 355 | 0.82 (0.63-1.06) | 5.15 |
| Poehling et al,[18] 2006 | 43 | 135 | 49 | 170 | 1.11 (0.79-1.56) | 2.98 |
| Doan et al,[22] 2009 | 16 | 89 | 23 | 110 | 0.86 (0.48-1.53) | 1.06 |
| Brendish et al,[21] 2017 | 101 | 133 | 113 | 146 | 0.98 (0.86-1.12) | 20.66 |
| May et al,[25] 2019 | 20 | 93 | 33 | 98 | 0.64 (0.40-1.03) | 1.53 |
| Bouzid et al,[16] 2021 | 160 | 275 | 115 | 199 | 1.01 (0.86-1.18) | 14.41 |
| Rao et al,[26] 2021 | 115 | 452 | 88 | 456 | 1.32 (1.03-1.68) | 5.79 |
| Bibby et al,[15] 2022 | 23 | 184 | 22 | 237 | 1.35 (0.78-2.34) | 1.14 |
| Mattila et al,[24] 2022 | 226 | 829 | 118 | 414 | 0.96 (0.79-1.16) | 9.76 |
| Random-effects model: $I^2$ = 0.03%; $\tau^2$ = 0 | | | | | 0.99 (0.93-1.05) | |




B RV positive vs negative test result

| Study | RV test Antibiotic | RV test Total | Control Antibiotic | Control Total | Risk ratio (95% CI) | Weight, % |
|---|---|---|---|---|---|---|
| Bonner et al,[20] 2003 | 7 | 96 | 27 | 97 | 0.26 (0.12-0.57) | 11.66 |
| Esposito et al,[23] 2003 | 14 | 43 | 282 | 435 | 0.50 (0.32-0.78) | 18.65 |
| Iyer et al,[17] 2006 | 15 | 105 | 73 | 240 | 0.47 (0.28-0.78) | 17.04 |
| Poehling et al,[18] 2006 | 4 | 28 | 39 | 107 | 0.39 (0.15-1.00) | 9.38 |
| Doan et al,[22] 2009 | 8 | 56 | 8 | 33 | 0.59 (0.24-1.42) | 10.18 |
| Brendish et al,[21] 2017 | 41 | 55 | 60 | 78 | 0.97 (0.80-1.18) | 23.75 |
| Bibby et al,[15] 2022 | 5 | 67 | 18 | 117 | 0.49 (0.19-1.25) | 9.34 |
| Random-effects model: $I^2$ = 65.69%; $\tau^2$ = 0.13 | | | | | 0.53 (0.37-0.77) | |

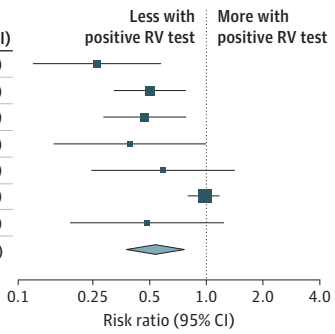



Table 3. Antibiotic Prescribing According to Rapid Viral Test Availability in Predefined Subgroups

| Category | Subgroup | No. of studies | No. of patients | Relative association estimate (risk ratio) | Absolute association estimate (risk difference) | Subgroup comparison |
|---|---|---|---|---|---|---|
| Age | Children and adolescents[a] | 9[15,17,18,20,22-26] | 5105 | 0.97 (95% CI, 0.83 to 1.12); $I^2$ = 54.5% | −0.01 (95% CI, −0.05 to 0.02); $I^2$ = 53.3% | $P$ = .82 |
| | Adults | 4[15,16,21,25] | 963 | 0.98 (95% CI, 0.89 to 1.09); $I^2$ = 0% | −0.01 (95% CI, −0.07 to 0.05); $I^2$ = 0% | |
| Test type | Traditional (antigen and immunofluorescence)[b] | 5[17,18,20,22,23] | 2552 | 0.91 (95% CI, 0.77 to 1.07); $I^2$ = 42.3% | −0.03 (95% CI, −0.07 to 0.01); $I^2$ = 18.4% | $P$ = .26 |
| | Molecular[c] | 6[15,16,21,24-26] | 3516 | 1.01 (95% CI, 0.92 to 1.12); $I^2$ = 20.6% | 0.01 (95% CI, −0.03 to 0.05); $I^2$ = 40.7% | |
| No. of targets | Monoplex (influenza) | 4[17,18,20,23] | 2353 | 0.91 (95% CI, 0.76 to 1.09); $I^2$ = 53.7% | −0.03 (95% CI, −0.08 to 0.02); $I^2$ = 35.0% | $P$ = .32 |
| | Multiplex (≥2) | 7[15,16,21,22,24-26] | 3715 | 1.01 (95% CI, 0.93 to 1.09); $I^2$ = 0.01% | 0.005 (95% CI, −0.03 to 0.04); $I^2$ = 35.5% | |
| Risk of bias | Low risk of bias | 7[20-26] | 4168 | 0.95 (95% CI, 0.82 to 1.10); $I^2$ = 67.1% | −0.02 (95% CI, −0.06 to 0.03); $I^2$ = 57.0% | $P$ = .73 |
| | High risk of bias | 4[15-18] | 1900 | 0.99 (95% CI, 0.87 to 1.12); $I^2$ = 5.5% | 0.0005 (95% CI, −0.04 to 0.05); $I^2$ = 29.8% | |
| Industry funding | None | 7[17,18,20-24] | 4074 | 0.97 (95% CI, 0.90 to 1.03); $I^2$ = 0% | −0.02 (95% CI, −0.05 to 0.00); $I^2$ = 0% | $P$ = .57 |
| | Industry funding | 4[15,16,25,26] | 1994 | 1.05 (95% CI, 0.79 to 1.39); $I^2$ = 70.2% | 0.01 (95% CI, −0.05 to 0.07); $I^2$ = 61.2% | |

[a] Cutoff differed between age 15 to 21 years according to the individual study.

[b] These studies were all published before 2010.

[c] These studies were all published since 2017.



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There was little or no difference between RV testing and the control in ED length of stay with moderate certainty (standardized mean difference, 0 hours, 95% CI, −0.17 to 0.16; $I^2$ = 67.4%; Table 2; eFigure 7 in Supplement 1). The follow-up interval for return ED visits varied between 7 and 30 days. Meta-analysis of 6 studies showed no difference in the number of return visits with moderate certainty (RR, 0.93; 95% CI, 0.79-1.08; $I^2$ = 0%; Table 2; eFigure 8 in Supplement 1). Nine studies investigated hospitalizations; meta-analysis demonstrated no association of RV testing with high certainty (RR, 1.01; 95% CI, 0.95-1.08; $I^2$ = 0%; Table 2; eFigure 9 in Supplement 1).[16-19,21,23-26]

Two studies assessed total costs of the ED stay, which were $33[17] and $173[24] higher per patient in the RV test arm. Rao et al[26] was the only study that surveyed the patient perspective: 21 of 314 families (7%) stated that the result of ED rapid multiplex testing contributed to how they subsequently sought medical care for their child's illness.

### Social Determinants of Health

All studies were from high-income countries. Overall data on social determinants of health were limited. All studies analyzed sex as a variable. Ethnicity was reported in 6 studies,[17,18,20,21,25,26] although 2 of these studies categorized the patients only as White or other.[17,21] Rao et al[26] was the only study to report data on additional social determinants of health: socioeconomic status, social capital, and insurance. Other domains according to the PROGRESS Plus framework,[7] namely place of residence, occupation, and religion, were not analyzed in any of the 12 studies.

## Discussion

In this systematic review and meta-analysis of RCTs, the availability of RV testing for respiratory viruses in EDs was not associated with overall antibiotic use. Fewer patients with a positive RV test result were prescribed antibiotics, counterbalanced by more prescribing for patients with a negative result. However, this was only observed in studies using monoplex antigen detection tests for influenza and not molecular or multiplex testing, suggesting that a rapid positive result for influenza was more likely to influence antibiotic prescribing than positive results for other respiratory viruses. RV testing was associated with higher antiviral use and a modest reduction in blood tests and chest radiography. There was no association with other outcomes evaluated, including overall costs. While study characteristics were heterogeneous, including pediatric and adult populations as well as monoplex and multiplex testing, results were mainly congruent.

These results align with those from previous systematic reviews. Lee et al[5] examined the effect of RV testing in ambulatory care among studies published to 2017 and also noted no association with antibiotic treatment among RCTs, but more antiviral use and fewer chest radiographs and blood tests. This meta-analysis included only traditional antigen detection-based rapid tests with limited sensitivity[27,28] and, in most cases, only 1 viral target. Our analysis included 6 additional RCTs that used highly sensitive multiplex molecular assays,[27,28] and despite recent advances in rapid test technology, had similar findings to Lee et al.[5] A 2023 systematic review and meta-analysis by Clark et al[29] focused on the association of multiplex panels in adults, mainly among hospitalized patients. Among RCTs in inpatients, there was no change in antibiotic prescriptions and a nonsignificant trend to shorter antibiotic duration. These RCTs found improved appropriateness of antiviral treatment and improved infection control practices, but no change in hospital length of stay. Both systematic reviews included RCTs as well as observational studies but only found significant reductions in antibiotic use in the latter. Several guidelines on viral testing and antibiotic stewardship refer to these observational studies, which are more susceptible to bias than RCTs due to selection and publication biases, confounding by indication, secular trends, and other sources of bias.[3,30,31] Accordingly, we believe that future recommendations should focus on the substantial evidence from the expanding number of RCTs.

Influenza antivirals were only given to 23.2% of influenza-positive patients in the RV testing arms, and rapid testing was associated with a pooled absolute RD of 1.4% in antiviral prescribing. Accordingly, the number needed to test for 1 additional antiviral prescription in these studies was approximately 70 (approximately 50 in adult studies and 100 in pediatric studies). Perhaps not surprisingly, given that most guidelines, including those of the Infectious Diseases Society of America,[32,33] recommend antiviral treatment only for patients early in the course of infection and for high-risk patients and/or severe or complicated disease, and that the benefits of outpatient antiviral therapy are limited,[34] clinicians in the included studies only prescribed antivirals to a minority of patients with influenza. Given the absence of benefit of RV testing on overall antibiotic use, these findings suggest that RV testing should not be routine, but rather should be reserved for patients for whom the testing will change treatment.[33] Current treatment guidelines for COVID-19 also only recommend antiviral treatment for high-risk patients and/or severe or complicated disease.[35] Considering that symptoms of influenza, COVID-19, and other respiratory infections can overlap, targeted multiplex viral testing in these patient populations should have greater clinical effect.

Our analysis of social determinants of health showed that these were generally not sufficiently evaluated and/or underreported. Importantly, all studies were from high-income countries. As antibiotic use is higher among marginalized communities within high-income countries and highest in middle-income countries,[36,37] the effect of viral testing might have been different in other patient populations, limiting the generalizability.

### Strengths and Limitations

This systematic review and meta-analysis had limitations. First, allocation concealment was not possible in any of the studies as effects work through awareness of the test result. Despite this, our main findings were consistent, including among the 7 studies considered to be low risk of bias. Second, information on antibiotic duration was limited. However, the 2 stud-



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ies which evaluated antibiotic duration did not show a difference between groups. Third, only 16% of all patients were adults. Additional RCTs in adults, especially those with high-risk conditions, would strengthen the evidence base. Nonetheless, a subgroup analysis of adults and children did not differ for our primary outcome. Fourth, there was uncertainty among some of the included studies as to whether the RV result was communicated before prescribing medications or ordering ancillary tests. This could bias results toward the null and underestimate the association of viral testing. However, this is a clinical reality in the ED where diagnostic testing and treatment decisions are made in parallel rather than sequentially. Finally, none of the studies were conducted since the start of the COVID-19 pandemic; therefore, none evaluated testing for SARS-CoV-2. However, as for influenza, testing for SARS-CoV-2 in EDs has been increasingly restricted to severe illness or high-risk patients for whom results would change treatment.[35]

A strength of our study is the focus on RCTs. We also used GRADE to systematically assess the certainty of evidence. Moreover, while individual studies had insufficient power to show some effects, pooling results from several studies allowed us to reveal these associations. Finally, we included data specific to ED from 2 studies that had not previously reported their ED data separately.[15,21]

## Conclusion

Overall, the results of this systematic review and meta-analysis suggest that the benefits of routine RV testing in the ED are limited. Such testing in EDs has no association with overall antibiotic use, length of ED stay, ED return visits, or hospitalization rates. Testing was associated with a minority of patients with influenza being prescribed antivirals and decreases in ordering of some ancillary tests. Patients with positive viral test results received less antibiotics compared with patients with negative test results, possibly improving the appropriateness of antibiotic treatment in this subgroup. Evidence suggests that RV testing in the ED should be reserved for patients for whom results will change treatment. Further RCTs in adults and high-risk populations are warranted.


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